

MICHAEL BLEHER, DR.

📍 Institute for Mathematics, Ruprecht-Karls-Universität Heidelberg, Germany.

✉ mbleher@mathi.uni-heidelberg.de

🆔 0000-0002-7796-1665

🌐 <https://michael.bleher.me>

ACADEMIC POSITIONS

2023–PRESENT **Postdoctoral Researcher**, *Heidelberg University*.

STRUCTURES Cluster of Excellence

Exploratory Project: New Methods for Single Cell Data Analysis – a Promising Future Direction for STRUCTURES?

Prof. Dr. Fred Hamprecht, Prof. Dr. Anna Marciniak-Czochra, Prof. Dr. Anna Wienhard.

ERC Project 'PERpetuating Stemness: From single-cell analysis to mechanistic spatio-temporal models of neural stem cell dynamics'.

Prof. Dr. Anna Marciniak-Czochra.

Research Directions:

- Geometry-preserving low-dimensional representations of scRNA-seq data.
- Neuron differentiation as a magnetic system on low-dimensional manifolds.
- Topological signatures for model selection in neuron differentiation dynamics.
- Topology of spatial transcriptome organisation.

EDUCATION

2017–2023 **PhD Physics and Mathematics (Interdisciplinary)**, *Heidelberg University*.

Advisor: Prof. Dr. Johannes Walcher

'Haydys-Witten Instantons and Symplectic Khovanov Homology'

2013–2016 **M.Sc. Physics**, *Heidelberg University*.

Advisor: Prof. Dr. Johannes Walcher

'A Survey of Defects in $\mathcal{N} = 4$ Supersymmetric Yang-Mills Theory'

2014–2015 **Graduate Course**, *University of Durham, UK* (ERASMUS+ Exchange).

M.Sc. in Particles, Strings and Cosmology at the Centre for Particle Theory.

2010–2013 **B.Sc. Physics**, *Heidelberg University*.

Advisor: Dr. Werner Rodejohann

'Neutrinoless Doppelbeta-Zerfall – Untersuchung einer Methode zur Auswahl eines Nuklearen Matrix-Elements'

SCHOLARSHIPS AND GRANTS

2025 **DAAD Conference Grant**, *German Academic Exchange Service (DAAD)*.

Support for attending the 'Geometric Realization of the AATRN', Chicago.

2017 - 2020 **Distinguished Doctoral Fellowship**, *Heidelberg Graduate School of Fundamental Physics*.

2014 - 2015 **ERASMUS+ Scholarship**, *European Commission*.

TEACHING

- 2022 **Mathematics Refresher for Master Students in Economics**, *Teaching Assistant, Prof. Dr. Thomas Dimpfl, University of Hohenheim* (available on YouTube).
- 2020–2021 **Helpdesk for Math Students**, *Teaching Assistant, Heidelberg University*.
- 2019 **Theoretische Physik IV**, *Teaching Assistant, Prof. Dr. Arthur Hebecker, Heidelberg University*.
- 2018 **Theoretische Physik I**, *Teaching Assistant, Prof. Dr. Tilman Plehn, Heidelberg University*.
- 2018 **Höhere Mathematik für Physiker III**, *Teaching Assistant, Prof. Dr. Hans Knüpfer, Heidelberg University*.
- 2018 **Theoretische Physik II**, *Teaching Assistant, Prof. Dr. Matthias Bartelmann, Heidelberg University*.
- 2017 **Applied Topology I**, *Teaching Assistant, Prof. Dr. Daniel Roggenkamp, University of Mannheim*.

SCIENTIFIC ENGAGEMENT

- 2026 **PMLR Volume 325: Geometry, Topology, and Machine Learning**, *Proceedings of Machine Learning Research*.
Editor, with F. Jensen, L. Maier, D. Taha, A. Wienhard.
- 2025 **Workshop on Geometry, Topology, and Machine Learning**, *MPI-MiS, Leipzig*.
Co-Organizer, with F. Jensen, L. Maier, D. Taha, A. Wienhard.
- 2025–PRESENT **Persistent Seminar**, *TDA Research Seminar, Heidelberg & MPI of Biochemistry*.
Co-Organizer, with F. Jensen (Heidelberg) and B. Stolz-Pretzer (MPIB).
- 2024–PRESENT **Tea, Coffee, Cake & TDA**, *Invited Speaker Series, Heidelberg University*.
Co-Organizer, with F. Jensen.
- 2024 **Dagstuhl Seminar 24122**, *Low-Dimensional Embeddings of High-Dimensional Data. Schloss Dagstuhl, Germany* (invited participant).
- 2023 **Workshop on Topological Methods in Data Analysis**, *Heidelberg University*.
Co-Organizer, with F. Jensen, V. Noel, A. Marciniak-Czochra.
- 2021 **Workshop on Topological Methods in Data Analysis**, *Heidelberg University*.
Co-Organizer, with D. Spitz, M. Schmahl, A. Wienhard.
- 2020 **Workshop on Topological Methods in Data Analysis**, *Heidelberg University*.
Co-Organizer, with D. Spitz, M. Schmahl, A. Wienhard.
- 2019–2023 **Topics in TDA**, *Journal Club, Heidelberg University*.
Co-Organizer, with F. Jensen, V. Noel, D. Spitz, M. Schmahl, A. Wienhard.
- Research Blog**, *Notes on TDA, geometric ML, and mathematical physics*.
subthaumic.substack.com

SOFTWARE DEVELOPMENT

- CHROMABENCH** **Chromatic TDA Benchmark**, *Python package*.
A benchmark for chromatic (label-aware) topological data analysis: labelled point clouds where colour-blind persistent homology provably cannot win.
<https://github.com/subthaumic/chromabench>.
- GNE** **geometric Neighbour Embeddings**, *Python package*.
Dimensionality reduction technique for embedding high-dimensional data while preserving local geometric relationships.
<https://github.com/subthaumic/gne>.
- MARKOVMODUS** **Markov-modulated unspliced/spliced count simulator**, *Python package*.
Generate synthetic scRNA-seq datasets where a hidden state graph (a continuous-time Markov process) modulates single cell unspliced/spliced RNA dynamics.
<https://github.com/subthaumic/markovmodus>
- MURIT** **Multiparameter Rips Transform**, *Go application*.
A Ripser add-on for computing persistent homology along custom subfiltrations of multi-filtered metric spaces.
<https://github.com/tdalife/murit>
- EVOTREC** **Evolutionary topological Recurrence**, *Python module*.
Implementation for computing the topological recurrence index of mutations in a given genome alignment.
<https://github.com/ottamj/evotrec>.

PUBLICATIONS

12. **Bleher, M.**, Jensen, F., Maier, L., Taha, D. and Wienhard, A. 'Preface of Geometry, Topology, and Machine Learning 2025'. In: *Proceedings of the Geometry, Topology, and Machine Learning Workshop*. Geometry, Topology, and Machine Learning. PMLR, 29th June 2026, pp. i–vii.
11. **Bleher, M.** 'A Family of Instanton-Invariants for Four-Manifolds and Their Relation to Khovanov Homology'. In: *SIGMA* (23rd Mar. 2026). DOI: 10.3842/SIGMA.2026.028. arXiv: 2412.13285 [math-ph].
10. **Bleher, M.**, Brazke, D. and Nill, S. (2026). *A Riemannian Autocorrelation Function and its Application to Non-Local Isoperimetric Energies*. arXiv: 2601.10481 [math] (preprint).
9. **Bleher, M.** (2025). *The tangled web they weave: Exploring neural networks with directed simplicial complexes* (in preparation).
8. Bodt, C. de, Diaz-Papkovich, A., **Bleher, M.**, Bunte, K., Coupette, C., Damrich, S., Sanmartin, E. F., Hamprecht, F. A., Horvát, E.-Á., Kohli, D., Krishnaswamy, S., Lee, J. A., Lelieveldt, B. P. F., McInnes, L., Nabney, I. T., Noichl, M., Poličar, P. G., Rieck, B., Wolf, G., Mishne, G. and Kobak, D. (2025). *Low-Dimensional Embeddings of High-Dimensional Data*. arXiv: 2508.15929 [cs] (preprint).
7. **Bleher, M.** (2025). *Adiabatic Solutions of the Haydys-Witten Equations and Symplectic Khovanov Homology*. arXiv: 2501.01365 [math-ph] (accepted at Advances in Theoretical and Mathematical Physics).
6. Bleher, J. and **Bleher, M.** (2024). *An Algebraic Framework for the Modeling of Limit Order Books*. arXiv: 2406.04969 [q-fin.TR] (preprint).
5. **Bleher, M.** (2023). *The Decoupled Haydys-Witten Equations and a Weitzenböck Formula*. arXiv: 2307.15056 [math.DG] (preprint).
4. **Bleher, M.** (2023). *Growth of the Higgs Field for Kapustin-Witten Solutions on ALE and ALF Gravitational Instantons*. arXiv: 2306.17017 [math.DG] (preprint).

3. Neumann, M., **Bleher, M.**, Hahn, L., Braun, S., Obermaier, H., Soysal, M., Caspart, R. and Ott, A. (2022). *MuRiT: Efficient Computation of Pathwise Persistence Barcodes in Multi-Filtered Flag Complexes via Vietoris-Rips Transformations*. arXiv: 2207.03394 [math.AT] (preprint).
2. **Bleher, M.**, Hahn, L., Patino-Galindo, J. A., Carriere, M., Bauer, U., Rabadan, R. and Ott, A. (2021). *Topology Identifies Emerging Adaptive Mutations in SARS-CoV-2*. arXiv: 2106.07292 [q-bio.PE] (preprint).
1. Bleher, J., **Bleher, M.** and Dimpfl, T. (2020). *From Orders to Prices: A Stochastic Description of the Limit Order Book to Forecast Intraday Returns*. arXiv: 2004.11953 [q-fin.TR] (preprint).

PRESENTATIONS

22. *Single Cells, Population Dynamics, and Euler Characteristic Profiles*. Division of Differential Equations and Applications of Mathematics Seminar, Gdansk University of Technology, Gdansk, Poland (invited talk). 28th May 2026.
21. *Single Cells, Population Dynamics, and Euler Characteristic Profiles*. MATRIX Workshop: Applications of Computational Geometry and Topology to Data Analysis of Complex Biological Data, Creswick, Australia (invited talk). 14th May 2026.
20. *Single Cells, Population Dynamics, and Euler Characteristic Profiles*. EPFL Topology Seminar, Lausanne, Switzerland (invited talk). 2nd Apr. 2026.
19. *The Tangled Web They Weave – Exploring Neural Networks with Directed Topology*. ATMCS 12, Leipzig, Germany (flash talk). 26th Jan. 2026.
18. *Population Dynamics, scRNA-seq, and Euler Characteristic Profiles*. Velten Lab Seminar, Heidelberg, Germany (invited talk). 20th Jan. 2026.
17. *Topological Signatures of Convergence in Viral Evolution*. Dioscuri TDA Seminar, Warsaw, Poland (invited talk). 12th Jan. 2026.
16. *Population Dynamics, scRNA-seq Data, and the Euler Characteristic Profile*. PEPS Seminar Heidelberg, 30th Oct. 2025.
15. *Haydys-Witten Instantons and the Gauge Theoretic Approach to Khovanov Homology*. Rutgers Gauge Theory Seminar, (invited talk). 14th Oct. 2025.
14. *The Tangled Web They Weave – Exploring Neural Networks with Directed Topology*. Geometric Realization of the AATRN, Chicago, USA (poster). 19th Aug. 2025.
13. *The Tangled Web They Weave – Exploring Neural Networks with Directed Topology*. Young Topologists Meeting 2025, Stockholm, Sweden (contributed talk). 25th June 2025.
12. *Harmonic Persistent Homology*. Persistent Seminar, Heidelberg, Germany. 28th May 2025.
11. *Persistence and Coarse-Graining in Dynamical Biological Systems*. TDA Symposium at MPI-MCG, Dresden, Germany (invited talk). 12th Dec. 2024.
10. *Fast Computation of Pathwise Persistence in Pandemic-Scale SARS-CoV-2 Genome Data*. 4th Workshop on Computational Persistence, TU Graz, Graz, Austria. 23rd Sept. 2024.
9. *Haydys-Witten Instantons and the Gauge Theoretic Approach to Khovanov Homology*. Gauge Theory and Mathematical Physics Seminar, Morningside Center of Mathematics, Beijing, China (invited talk). 3rd July 2024.
8. *RNA Velocity Embeddings in Curved Spaces - Exploring Cellular Dynamics*. Seminar 24122, Dagstuhl, Germany. 20th Mar. 2024.
7. *On Haydys-Witten Instantons and the Gauge Theoretic Approach to Khovanov Homology*. HU Gauge Theory Research Seminar, Berlin, Germany (invited talk). 31st Jan. 2024.

6. *Haydys-Witten Instantons in the Gauge Theoretic Approach to Khovanov Homology*. ULB Geometry Seminar, Brussels, Belgium (invited talk). 4th Dec. 2023.
5. *Topological Signatures of Convergence in Viral Evolution*. CompTopNN Meeting 2023, Seville, Spain (invited talk). 8th Nov. 2023.
4. *Feature Representation of scRNA Data in Symmetric Spaces*. Structures Symposium, Heidelberg, Germany (poster). 20th July 2023.
3. *Persistent Homology Detects Emerging Adaptive Mutations*. TDA Journal Club, Heidelberg, Germany. 7th June 2021.
2. *Welcome Notes and an Introduction to Mapper*. Heidelberg TDA Workshop 2020, Heidelberg, Germany (organizer). 26th Oct. 2020.
1. *Nahm Pole Boundary Conditions and Lagrangian Intersection Floer Homology*. Quantum Fields, Knots, and Strings, Warsaw, Poland (poster). 2018.